GUJARAT TECHNOLOGICAL UNIVERSITY
AHMEDABAD INSTITUTE OF TECHNOLOGY

Report on:

Develop an Android application to register mobile shop customers and maintain their details using Firebase

B.E.: Semester-3 (Computer Engineering)

Subject: Database Management System (BE03000091)

Submitted by:

Name of student: GADHAVI INDRAJIT V.

Enrolment number: 250020107020

HOD: HARSHA PADHERIYA

Faculty: Khusharth Oza

Mobile Shop Customer Manager

Application Review Report (Android + Firebase)

Platform: Android | Backend: Google Firebase | Date: 9 November 2026

1. Executive Summary

The Mobile Shop Customer Manager is an Android application that lets shop staff register walk-in customers and keep their personal and purchase records in Google Firebase. It replaces paper registers and scattered phone contacts with one searchable, cloud-based record that updates on every device in real time.

Staff sign in with Firebase Authentication, add a customer in under a minute, and look up any customer by name or mobile number. Each record stores contact details, address, the device purchased (brand, model, IMEI), purchase date, warranty expiry and notes. Data lives in Cloud Firestore, so it stays safe if the phone is lost or changed.

This report covers the structure, features, database design and design of the application. It is a design and review document: no live shop figures are included.

2. Application Overview

Item	Detail
Name	Mobile Shop Customer Manager (Android)
Purpose	Customer registration and record keeping for a mobile phone retail shop
Users	Shop owner and staff (roles: Admin, Staff)
Platform	Android 8.0 (API 26) and above
Language	Kotlin (or Java) built in Android Studio
Backend	Google Firebase: Authentication, Cloud Firestore, Cloud Storage
Currency	Indian Rupee (₹)
Status	Offline caching enabled; data syncs when the phone is online

3. Structure and Features

3.1 Login and Home Dashboard

Staff sign in with email and password through Firebase Authentication. The home screen then gives a quick view of the shop:

- Summary cards: Total Customers, Registered Today, Warranty Expiring Soon, and Total Sales Value.
- Recent Customers list with name, mobile number, city and registration date.
- Quick buttons to add a customer or open the search screen.

3.2 Customer Registration

- Form fields: full name, mobile number, email, address, city, state and postal code.

- Purchase fields: brand, model, IMEI number, price in ₹, purchase date and warranty period.
- Optional customer photo and bill image saved in Firebase Cloud Storage.
- Input checks: 10-digit mobile number, valid email, and a duplicate-mobile warning before saving.

3.3 Customer Management

- Search by name or mobile number, with filters for Today, Past 7 Days and Past 30 Days.
- Sort by newest, oldest, name A to Z and name Z to A.
- Customer profile screen with full details and purchase history.
- Edit and delete a record (delete limited to Admin).
- Tap-to-call and tap-to-WhatsApp buttons for follow-up.
- Export of the customer list to CSV.

3.4 Firebase Database Design

Collection	Main fields
users	uid, name, email, role (Admin or Staff)
customers	customerId, name, mobile, email, address, city, state, pincode, notes, createdAt, createdBy
purchases	purchaseId, customerId, brand, model, imei, price, purchaseDate, warrantyExpiry
activityLogs	logId, userId, action, targetId, timestamp

Customers and purchases are linked by customerId. Firestore security rules allow only signed-in users to read or write, and only Admin users to delete.

4. Strengths

- Simple three-screen flow (Home, Register, Customers) that new staff can learn quickly.
- Real-time sync: a customer added on one phone shows on every other phone at once.
- Offline support: records can be added without internet and upload when the connection returns.
- No server to buy or maintain; Firebase handles hosting, scaling and backup.
- Secure sign-in and role-based rules protect customer data.
- Warranty and purchase details support after-sales service and repeat business.
- Search, filters, sorting and CSV export are built in.
- Activity log gives an audit trail of who changed what.

5. Observations and Risks

Area	Observation
Data privacy	Names, mobile numbers and addresses are personal data. Weak Firestore rules or shared logins could expose them.

Area	Observation
Internet dependence	Offline entries sync later; a long offline period can cause edit conflicts.
Firebase cost	The free Spark plan has daily read and write limits. A busy shop may need the paid Blaze plan.
Duplicate records	The same customer may be entered twice with different spellings unless mobile numbers are checked.
Inventory	There is no stock or product catalogue, so stock levels cannot be tracked.
Billing	No invoice, payment status or EMI tracking is included.
Reporting	Totals are lifetime only. No monthly sales trend or best-selling brand view.
Backup	Firestore has no automatic export schedule unless one is set up.

6. Recommendations

- Write strict Firestore security rules and give each staff member a separate login.
- Check for an existing mobile number before saving, and show a clear error or empty-state message when data fails to load.
- Add reports: sales by month, top-selling brands, warranty expiry list, and customer growth by city.
- Add an inventory module for phones and accessories, with low-stock alerts.
- Add invoice generation, payment and EMI tracking, and repair or service records.
- Send warranty-expiry and festival-offer reminders through Firebase Cloud Messaging.
- Add PDF and Excel export, and schedule regular Firestore backups.

7. Next Steps

To turn this design into a working project with real figures, build and test the app with sample data, then connect it to a Firebase project. Share a CSV export of the customers and purchases collections, or screenshots of the running app with data visible. The report can then include customer growth, sales by brand, warranty status and screenshots of each screen.